

Title

Simvastatin as an intracanal medicament compared with hydroxylated-calcium standard in symptomatic periapical inflammation: a randomized trial of pain and immune modulation

Abstract

Periapical inflammation following root canal preparation causes pain and progressive tissue destruction. Calcium-based medications remain standard practice, though alternative agents with immunological properties may offer complementary benefits. We compared a statin-based gel formulation with the conventional hydroxylated-calcium standard as intracanal medication in 30 patients with confirmed necrotic premolars and acute apical inflammatory symptoms. Participants were randomly assigned (n=15 per group) to receive either statin gel or hydroxylated-calcium following mechanical instrumentation. Pain intensity was recorded using numerical rating scales at 6, 12, 24, and 48 hours post-treatment. Inflammatory markers in periapical exudate were quantified at baseline and after one week. At 6 hours, statin-treated patients reported lower pain scores compared to hydroxylated-calcium controls. Pain ratings converged between groups by 24 hours post-treatment. Both medications reduced inflammatory markers from baseline; statin treatment demonstrated differential effects on different inflammatory mediators. No adverse events were documented in either group. While both medications proved clinically effective, preliminary findings suggest statin-based intracanal formulations warrant further investigation as potentially useful additions to standard endodontic protocols.